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Potential Brazilian wood species for bows of string instruments

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Abstract

Caesalpinia echinata Lam. (pau-brasil) is a Brazilian tree with symbolic character for Brazilian forestry, which is at risk of extinction because of illegal logging activities and exportation as raw material for bows of string instruments. Additional studies are required to select substitutes for pau-brasil, i.e., other wood species are being sought with potentially suitable characteristics for bow manufacture. The present paper is dedicated to the evaluation of six native Brazilian woods by comparing their anatomical features, specific gravity, speed of sound propagation, and dynamic modulus of elasticity in relation to the corresponding properties of *C. echinata* that are considered as standard for bow manufacture. The results showed that length, diameter, lumen diameter, and wall thickness of fibers are the most important variables contributing to specific gravity. Additionally, ray dimensions influenced sonic velocity. The values of dynamic modulus of elasticity are directly related to dimensions of rays and fibers. Among the six woods included in this study, *Dipteryx* spp., *Handroanthus* spp., and *Hymenaea* spp. were considered as the best alternatives to *C. echinata* wood as sources of bow construction. The bows from the first two woods were already tested by professional musicians and their suitability was certified. The *Diplotropis* spp. wood also showed promising characteristics. *Mezilaurus itauba* and *Astronium lecointei* do not have the necessary properties for bows.

Keywords: bows for string instruments; *Caesalpinia echinata*; potential woods for bow manufacture; wood properties.

Introduction

There are records of bows related to music from pre-history, as groundbow and earth-bow, both from Africa. The Chinese violin made of bamboo, with bows of the same material, is still used in Chinese traditional music. Violins from Hindu

and Greek mythology used bows that did not have curvatures (Baines 1961).

In Europe, the bows of the violin family became known around the 10th century along with Arab invasions. From that period until the middle ages and early renaissance, these instruments have been changed and improved gradually in accordance with the requirements of music. During the 16th and 17th centuries, instruments and bows have been further developed driven by new musical trends. The curvature of bows changed from concave to convex, the mechanism of horsehair's tensioning was improved, and size and weight of bows was optimized (Baines 1961; Rymer 2004; Alves 2008a).

Before the middle of the 18th century, various woods were employed in the manufacture of bows. The predominant species were from South and Central America because of their high density and stiffness. *Brosimum guianense* and *Zollernia paraensis* were popular during the baroque period, and *Caesalpinia echinata* became dominant during the classic-romantic and modern periods (Baines 1961; Holz 1996; Rymer 2004; Brémaud et al. 2008; Alves 2008a).

In the second half of the 18th century, Francois Tourte established that *Caesalpinia echinata* wood was ideal for bows for the classical-modern violin models (Baines 1961; Lucchi 2009), and it is still the most accepted wood by bow makers and musicians owing to the following reasons: high density, facility to be bent, stability of the curvature, durability, and beauty (Bueno 2002; Pierce 2002). However, as stated by the IBAMA (Brazilian Environment Institute) resolution 37-N, from April 1992, *C. echinata* is endangered (Rocha and Simabukuro 2008; Gasson et al. 2009) and its legal commercial use is restricted beginning from 2007. Therefore, studies on alternative wood species in manufacturing of bows became necessary. The choice of other woods should rely on specific characteristics such as specific gravity above 950 kg m⁻³ and speed of sound propagation equal or greater than 5000 m s⁻¹ (Longui 2005; Wegst 2006; Alves et al. 2008b). Several authors including Kollmann and Côté (1968), Panshin and De Zeeuw (1964), and Hoadley (2000) stated that density plays a key role and also governs other wood properties. Slooten and van der Souza (1993) described the sound propagation as being important in the manufacture of wooden musical instruments. Franco and Yojo (2008) emphasized that a high elasticity modulus is an important feature for construction of bows.

Manilkara spp. and *Handroanthus* spp. woods have been tested in Brazil for the production of modern bows (Longui 2005). It was concluded that *Handroanthus* spp. has similar or superior physical, mechanical, and acoustic properties as *C. echinata* wood. However, *Handroanthus* spp. woods are yellow, contrasting to the traditional red color of *C. echinata*

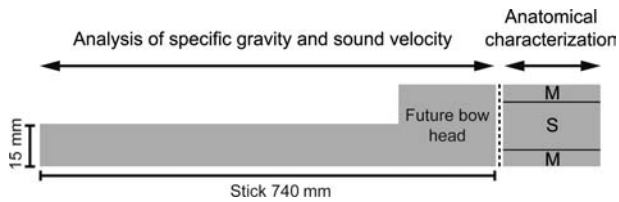


Figure 1 Schematic illustration of samples for anatomical analysis, specific gravity, and speed of sound propagation. M=macerations, S=sections.

wood preferred by musicians. Wegst et al. (2007) also noted the appearance of a bow is essential for the acceptance by musicians.

The search for alternative woods has to take into consideration anatomical features, physical, mechanical, acoustical, and chemical properties, in addition to workability to bows. Subsequently, the bows should be tested by musicians. The present study aims at searching for alternative woods for bow production, if possible for qualities suited for soloists, but at least for amateurs and beginners if the wood is cheaper than that of *C. echinata*.

The following woods were investigated: *Handroanthus* spp., *Hymenaea* spp., *Mezilaurus itauba*, *Dipteryx* spp., *Diploptropis* spp., and *Astronium lecontei*. The parameters specific gravity, speed of sound propagation, and dynamic modulus of elasticity should be tested and the relationships between anatomical features and wood properties estab-

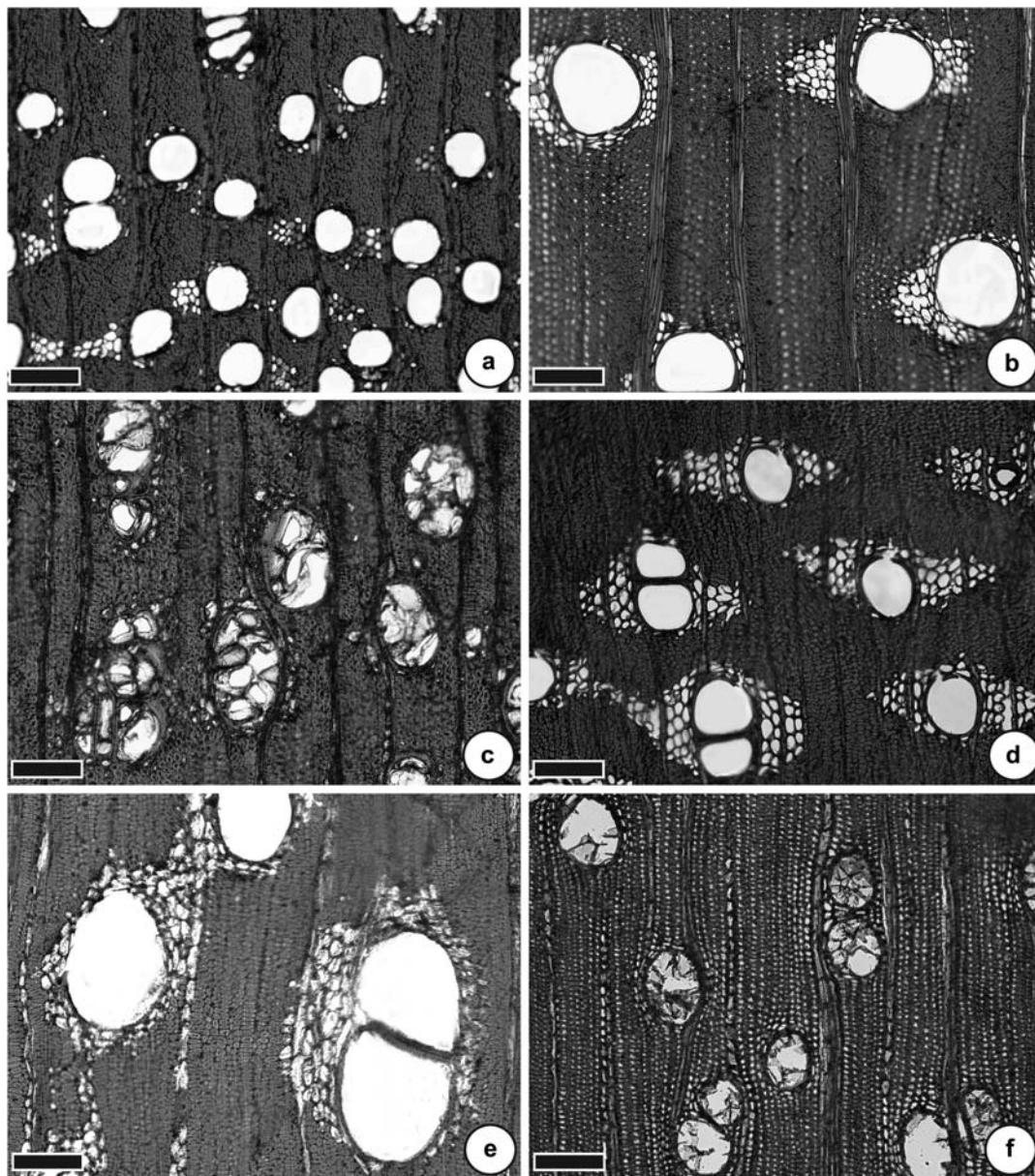


Figure 2 Transverse sections; note the difference in vessel diameter and vessel frequency among the wood species. (a) *Handroanthus* spp. (b) *Hymenaea* spp. (c) *Mezilaurus itauba*. (d) *Dipteryx* spp. (e) *Diploptropis* spp. (f) *Astronium lecontei*. Scale bars = 200 μ m.

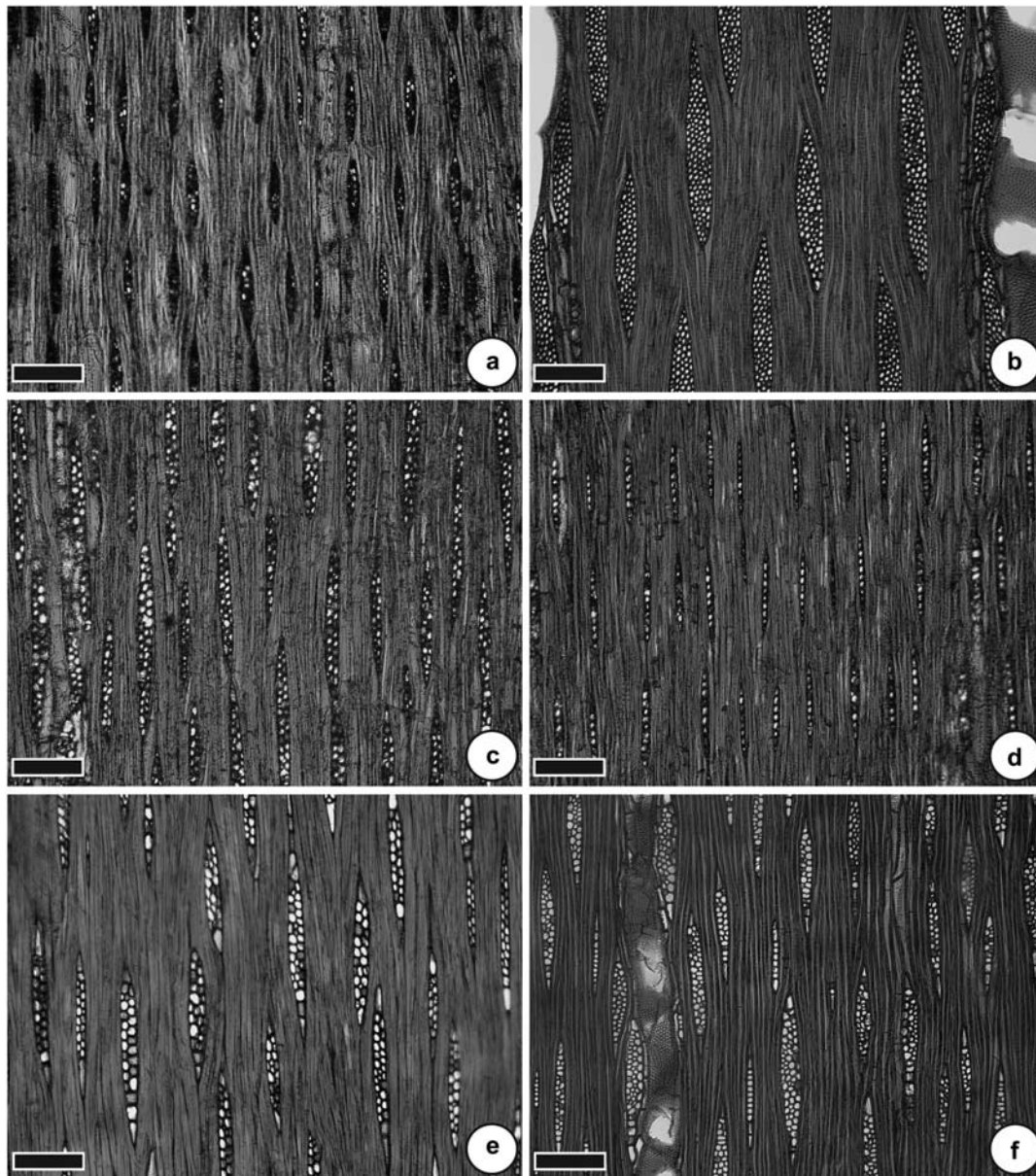


Figure 3 Longitudinal sections showing variation in ray dimensions and frequency. (a) *Handroanthus* spp. (b) *Hymenaea* spp. (c) *Mezilaurus itauba* (d) *Dipteryx* spp. (e) *Diplotropis* spp. (f) *Astronium lecointei*. Scale bars = 200 μm .

lished. The results will be discussed in relation to *C. echinata* wood, which is as a type of reference in the manufacture of bows for stringed instruments.

Materials and methods

Sample selection

Planks of *Handroanthus* spp. (ipê), *Mezilaurus itauba* (itaúba), *Hymenaea* spp. (jatobá), *Dipteryx* spp. (cumarú), *Diplotropis* spp. (sucupira), and *Astronium lecointei* (muiracatiara) were bought in the wood market in São Paulo. According to Mainieri et al. (1983), these wood species have specific gravity above 950 kg m^{-3} , a density considered to be ideal for bows (Longui 2005; Alves et al.

2008). To ensure that the samples were from different trees, only one plank was purchased in each woodworking supply store. The woods were then compared with samples from the Forestry Institute Xylarium (SPSFw).

Five planks of *Handroanthus* spp. and three planks from each of the other wood species were evaluated; sticks were cut ($740 \text{ mm} \times 15 \text{ mm} \times 15 \text{ mm}$) from these planks for the manufacture of bows. In total, 15 sticks of *Handroanthus* spp. were prepared, 3 from each plank and 6 from each one of the other types of wood, 2 of each planks; altogether 30 sticks were studied.

Anatomical analysis

Blocks of 1.5 cm^3 were cut from one end of the sticks (Figure 1). The blocks were softened in boiling water and glycerin (4:1), and $15\text{--}20 \text{ }\mu\text{m}$ thick sections were prepared on a sliding microtome.

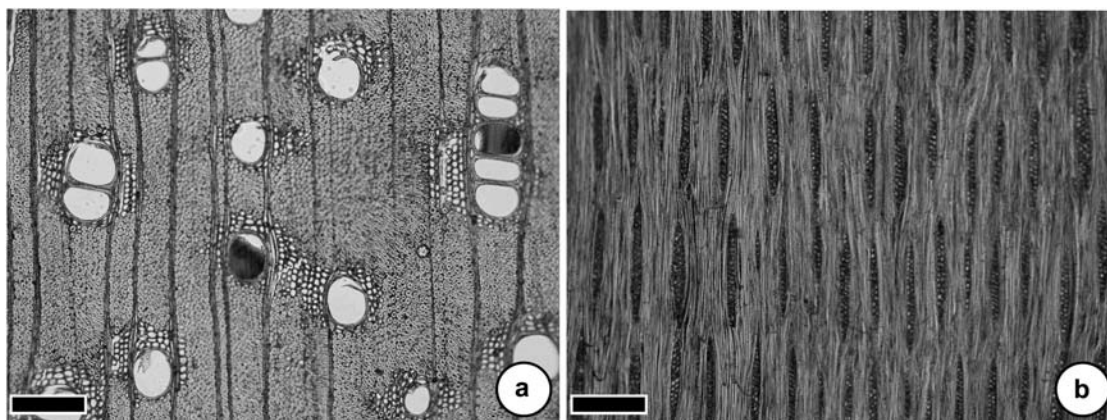


Figure 4 *Caesalpinia echinata* wood. (a) Transverse section. (b) Longitudinal sections. Compare with Figures 2 and 3. Scale bars = 200 μm .

The transverse and longitudinal sections were bleached with sodium hypochlorite (60%), washed in water, and stained with safranin (1%) (Johansen 1940). Macerations were prepared according to the modified Franklin method (Berlyn and Miksche 1976), stained with aqueous safranin and mounted in a solution of water and glycerin (1:1). The terminology followed the IAWA list (IAWA Committee 1989). All anatomical measures were performed on a microscope (Olympus CX 31) equipped with a camera (Olympus Evolt E330) and a computer with image analyzer software (Image-Pro 6.3).

Specific gravity (ρ_{ap})

Specific gravity was determined from the relation between the stick mass and volume of water displaced when it was immersed in a graduate cylinder containing water. Calculation: $\rho_{ap} = P_u / V_u$, where ρ_{ap} = specific gravity (kg m^{-3}), P_u = oven-dry weight of wood (kg), V_u = dry volume of wood (m^3).

Speed of sound propagation (v)

The speed sound propagation (v) was determined by the G. Lucchi Elasticity Tester, the instrument, which is applied by luthier and bow makers around the world. The tester emits an ultrasonic pulse from a transducer placed in contact with one end of the stick, and the pulse is detected by a receiver placed at the other end. The stick length was determined with measuring tape. Then, the speed of sound propagation (v) according to Lucchi (1986) was calculated: $v = m/s$, where v = speed of sound propagation (m s^{-1}), m = stick length (meters), s = time for the ultrasonic pulse to pass through the stick (seconds).

Dynamic modulus of elasticity (MOE_d)

Considering the values of specific gravity and speed of sound propagation, the dynamic modulus of elasticity was calculated by the formula: $\text{MOE}_d = (v^2/100) \times \rho_{ap}$, where MOE_d = dynamic modulus of elasticity (MPa), v = velocity (m s^{-1}), ρ_{ap} = specific gravity (kg m^{-3}).

Statistical analysis

Statistical analysis was performed employing the SigmaStat 3.5 software for Windows (SPSS Incorporation). For the comparison of each variable among different wood species, analysis of variance was applied according to the data distribution. (1) For normal dis-

tribution: parametric statistics (one way analysis of variance) was used, and the results were expressed as mean and standard deviation. (2) For non-normal distribution: non-parametric statistics (Kruskal-Wallis one way analysis of variance on ranks) were applied, and the values were expressed as median and p25–p75. (3) In the case of a significant difference: the Tukey or Dunn test was applied to identify the pair determinants of differences. The correlation among different variables was performed by linear regression. The results of the six woods were also compared with those of *Caesalpinia echinata*, considered as the reference wood for classical-modern bow manufacture.

Results

Comparisons and correlations between anatomical features and wood properties

A comparison between anatomical features and properties is given in Figures 2–8. Data of *Caesalpinia echinata* wood as reference are also presented (Alves et al. 2008b). *Handroanthus* spp., *Diplotropis* spp., *Dipteryx* spp., and *Hymenaea* spp. have axial parenchyma of the type aliform paratracheal; *Dipteryx* spp. lozenge-aliform; *Hymenaea* spp. with marginal bands; *Mezilaurus itauba* and *Astronium lecointei* scanty paratracheal. Regression analyses revealed that the characteristics of the fibers have a major impact on specific gravity, but are also impacted by ray characteristics. The ray dimension influences the speed of sound propagation, which is also influenced by the fiber lumen diameter and fiber wall thickness. Diameter and lumen of fibers influence the MOE_d , which is related to wood rigidity. The vessels length also showed correlation with these properties (Table 1). There is no correlation between the density and the speed of sound propagation.

Discussion

Fibers, lengthy cells with axial orientation, are the major component of wood and influence essentially its properties, especially its density (Fujiwara et al. 1991). As demonstrated

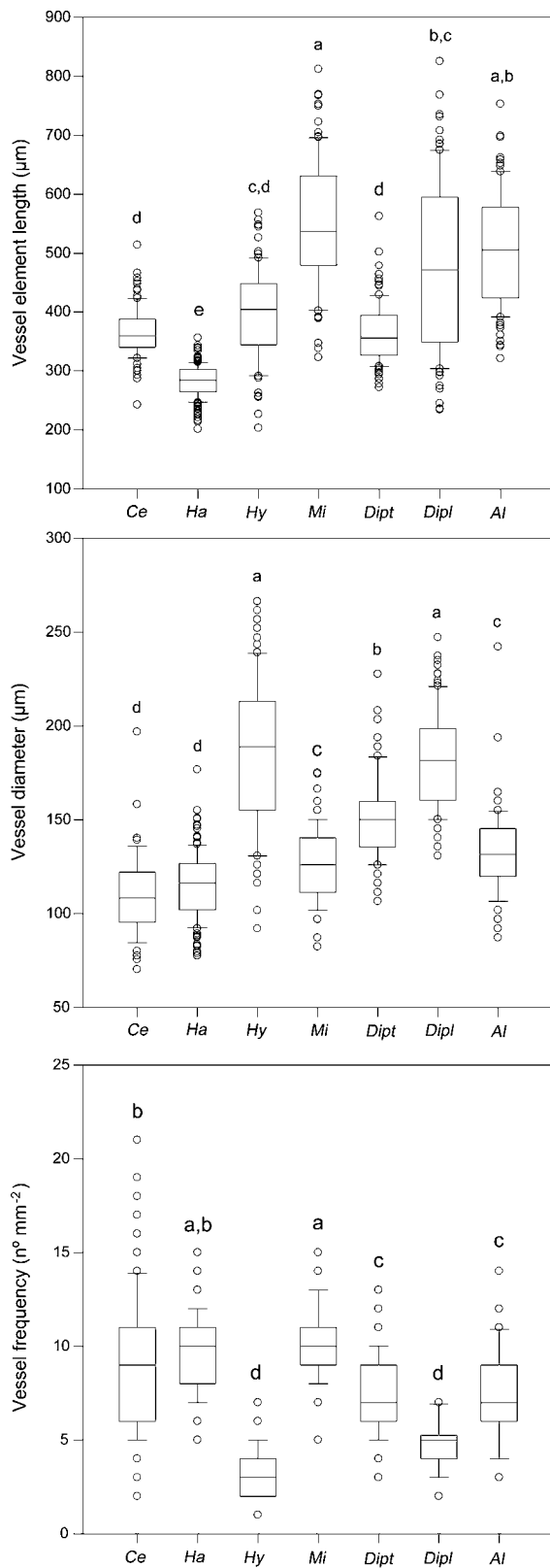


Figure 5 Vessel features for six potential wood species for bows and *C. echinata*.

Ce=*C. echinata*, *Ha*=*Handroanthus* spp., *Hy*=*Hymenaea* spp., *Mi*=*Mezilaurus itauba*, *Dipt*=*Dipteryx* spp., *Dipl*=*Diploptropis* spp., *Al*=*Astronium lecontei*. Distinct letters differ statistically at $P < 0.05$ with the Dunn test. *C. echinata* data are from Longui (2005).

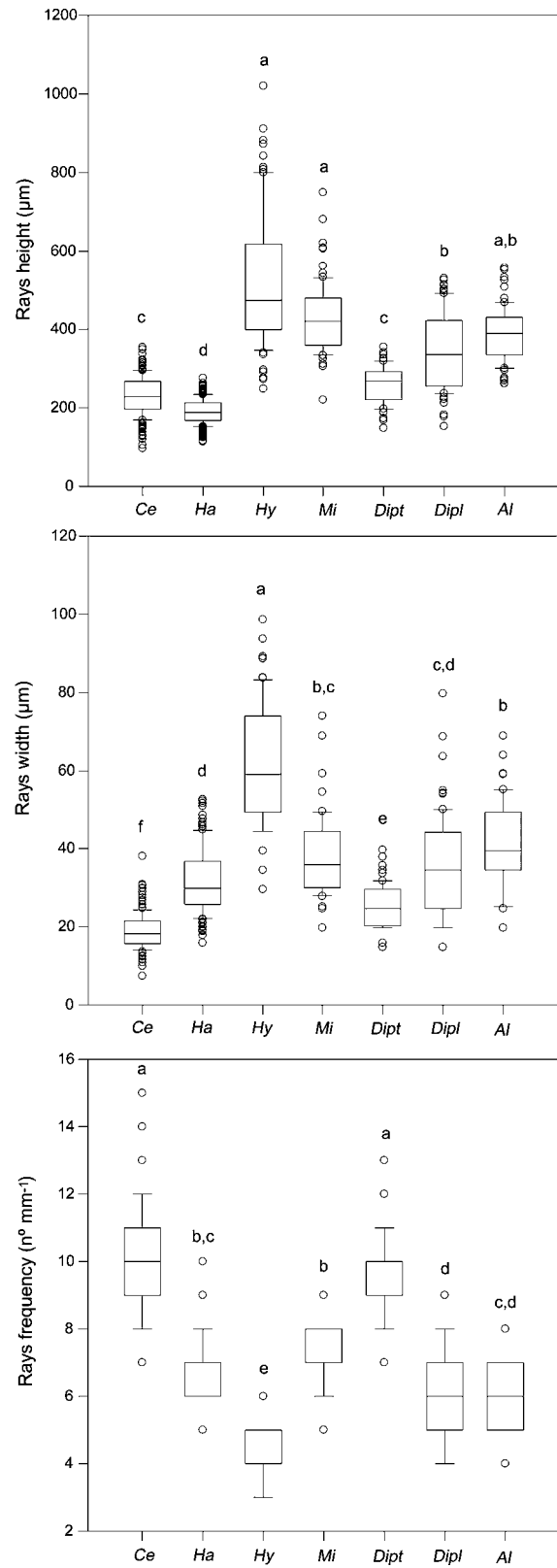


Figure 6 Ray features for six potential wood species for bows and *C. echinata*.

For abbreviations, see Figure 5. Distinct letters differ statistically at $P < 0.05$ with the Dunn test. *C. echinata* data are from Longui (2005).

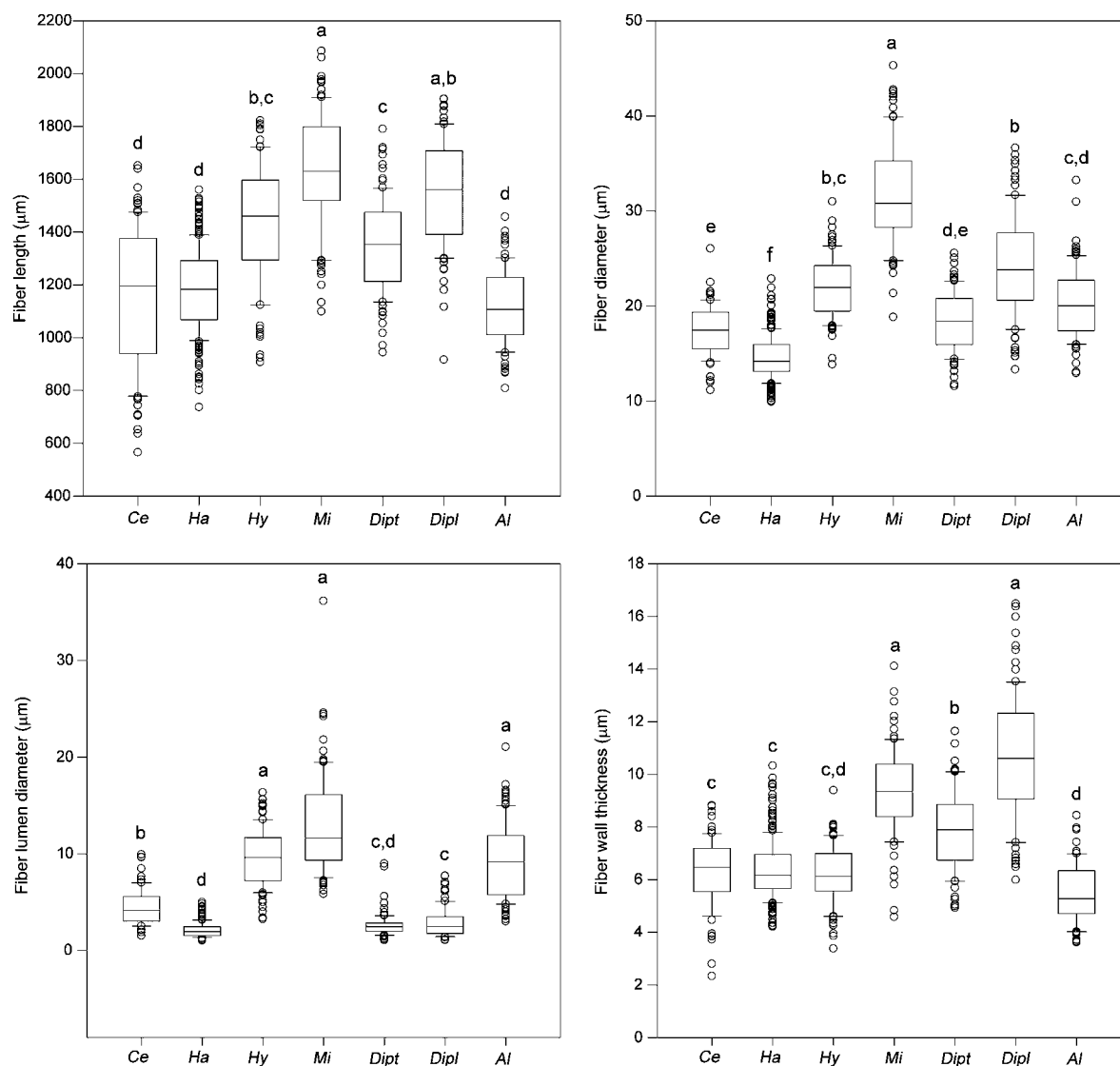


Figure 7 Fiber features for six potential wood species for bows and *C. echinata*.

For abbreviations, see Figure 5. Distinct letters differ statistically at $P < 0.05$ with the Dunn test. *C. echinata* data are from Longui (2005).

in Table 1, fiber characteristics, such as length, diameter, lumen diameter, and wall thickness, show a multitude of correlations with specific gravity. In *Dipteryx* spp., a positive correlation was observed between fiber wall thickness and specific gravity in agreement with reports by Denne and Hale (1999), Longui (2005), and Quilhó et al. (2006). A positive correlation was also observed between specific gravity and fiber length in *Dipteryx* spp. and *A. lecoinei*. Similar results were presented by Shimoyama (1990), Butterfield et al. (1993), and Longui (2005).

According to Wegst (2006), Wegst et al. (2007), Alves et al. (2008b), and Schimleck et al. (2009), density is an important property in the selection of wood for musical instruments and bows. Because bows for violins have a pre-determined weight, between 60 and 64 g (Alves et al. 2008b), their construction must be thinner in the case of sticks with higher density. In this study, the specific gravity of *M. itauba* and

A. lecoinei sticks was less appropriate for the manufacture of bows (Figure 8).

The cellular composition also influences the speed of sound propagation. Although the bows do not produce sound, the acoustic quality of wood can be considered as a parameter to qualify sticks for bows (Matsunaga et al. 1996). Brancheriau et al. (2006a,b) reported that woods for xylophone construction have a better sound quality if the axial parenchyma are of the paratracheal types. The same result was reported by Pourtahmasi and Golpayegani (2009) who investigated *Morus alba* wood in the manufacture of the “Tar”, an Iranian stringed instrument. In this study, all wood species investigated had different types of paratracheal axial parenchyma. *C. echinata* has axial parenchyma of the type aliform paratracheal (Figure 4). The question, which of them would have greater contribution to speed of sound propagation, could not be clarified because *A. lecoinei* and *M.*

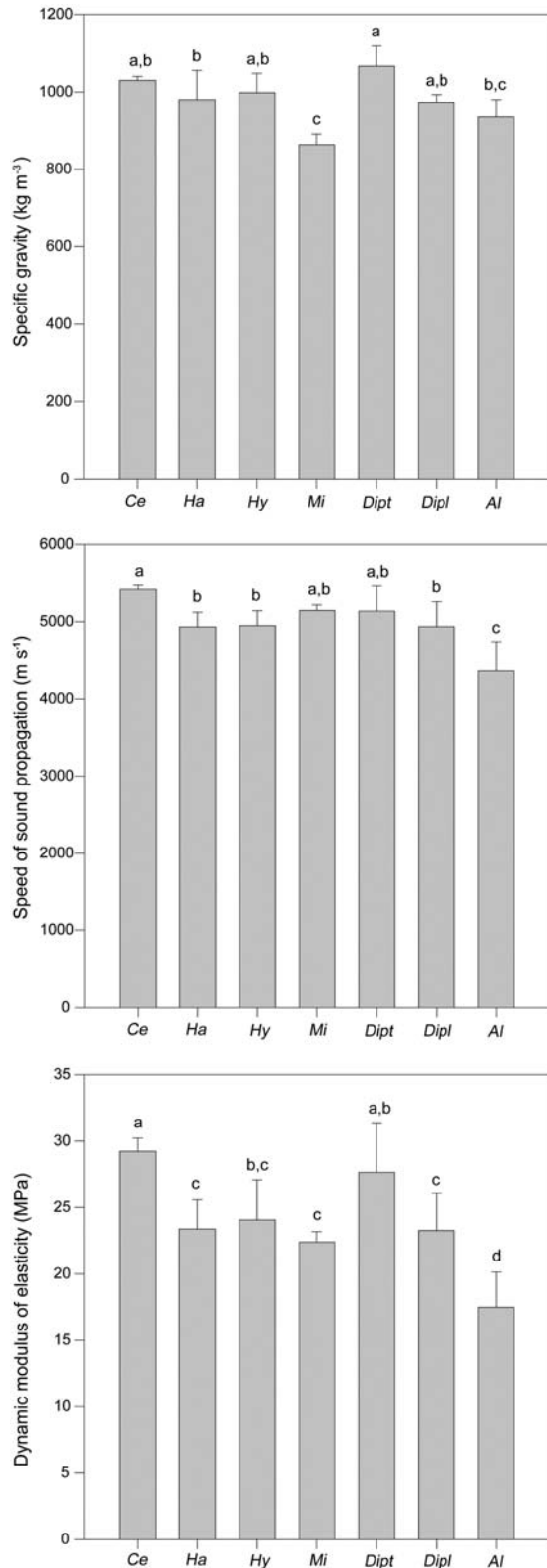


Figure 8 Specific gravity, speed of sound propagation, and dynamic modulus of elasticity for six potential wood species for bows and *C. echinata*.

For abbreviations, see Figure 5. Distinct letters differ statistically at $P < 0.05$ by the Tukey test. *C. echinata* data are from Longui (2005).

itauba, both with axial parenchyma of the type scanty paratracheal, revealed a high speed and a low speed, respectively. *Dipteryx* spp. with axial parenchyma of the type lozenge-aliform had the same sound speed as *M. itauba*. Brancheriau et al. (2006a,b) highlighted the importance of the rays in the acoustic quality of wood and noted that woods with short, homogeneous, and rays with low frequency had optimal acoustic quality for xylophones. Longui (2005) observed that *C. echinata* sticks with larger rays had a lower speed of sound propagation despite their higher specific gravity. In our study, larger and more frequent rays were related to lower speed of sound propagation in *Handroanthus* spp., *Dipteryx* spp., and *Diploptropis* spp. Rays are proceeding in radial direction, and theoretically they should slow down the sound wave propagation. In addition, very large rays can cause a deviation of adjacent fibers from the axial direction, and this fact contributes to reduction of sound speed.

In addition to axial parenchyma and rays, vessel features also influence the acoustic quality of wood. Alves et al. (2008b) found a negative correlation between vessels frequency and speed of sound propagation in *C. echinata* sticks. In this study, wood species with smaller vessel diameter, as *Handroanthus* spp. showed lower sonic velocity than those with larger but less frequent vessels, such as *Hymenaea* spp., Brancheriau et al. (2006a,b) observed a positive correlation between storied ray structure and sonic speed, whereas Longui (2005) and Alves et al. (2008b) did not found any relationship between these parameters in *Handroanthus* sp. and some samples of *C. echinata*. Among the wood species examined in this study, *Handroanthus* spp. *Dipteryx* spp. had storied ray structure, but other wood species without this feature showed higher sound of speed.

According to some authors, wood species with high sonic velocity usually have a high proportion of longer fibers with thicker walls (Bucur 1988; Bucur et al. 2002; Huang et al. 2003; Oliveira and Sales 2006; Pourtahmasi and Golpayegani 2009). In this study, the relation between fiber dimensions and the sound velocity was observed in *Diploptropis* spp. and *A. lecointei*. In the former, there was a negative correlation with the fiber diameter, and in the latter the correlation was positive with the fiber wall thickness.

A positive correlation between acoustic velocity and density was previously reported by Feeney et al. (1998), whereas other authors found negative relationships (Lucchi 1986; Fernandes 2004; Calegari et al. 2007). In the present study, no correlation between specific gravity and acoustic velocity was observed. Wang et al. (2003) made the same observation on *Taiwania cryptomerioides* wood. Likewise, Wegst (2006) found high sonic velocity both in wood species with high density as *C. echinata* and *Manilkara* spp., and in those with low density such as *Pinus sylvestris* or very low density as *Ochroma* spp.

M. itauba and *Dipteryx* spp. have significantly different specific gravities (860 and 1070 kg m⁻³, respectively) and stiffnesses (*Dipteryx* spp.: 27 663 MPa and *M. itauba*: 22 395 MPa). Therefore, stiffness seems to be a more important property in determining the quality of stick bows than gravity. The findings of Carrasco and Azevedo Júnior (2003)

Table 1 Regression equations showing the correlation between anatomical features and wood properties.

Wood	Equation	F	P	R ²
<i>Handroanthus</i> spp.				
	$\rho_{ap} = -105.608 + (73.802 \times \text{FD})$	10.800	0.006	0.45
	$v = 5432.699 - (15.960 \times \text{RW})$	6.387	0.025	0.32
<i>Mezilaurus itauba</i>				
	$\rho_{ap} = 1652.623 - (109.598 \times \text{RF})$	15.948	0.016	0.79
	$\rho_{ap} = 1055.267 - (15.016 \times \text{FLD})$	8.511	0.043	0.68
	$\text{MOE}_d = 33.575 - (0.356 \times \text{FD})$	27.601	0.006	0.87
	$\text{MOE}_d = 28.480 - (0.476 \times \text{FL})$	19.776	0.011	0.83
<i>Dipteryx</i> spp.				
	$\rho_{ap} = 757.886 + (0.877 \times \text{VEL})$	8.132	0.046	0.67
	$\rho_{ap} = 744.513 + (1.235 \times \text{RH})$	50.773	0.002	0.92
	$\rho_{ap} = 588.345 + (19.049 \times \text{RW})$	52.056	0.002	0.92
	$\rho_{ap} = 563.955 + (0.374 \times \text{FL})$	10.864	0.030	0.73
	$\rho_{ap} = 575.764 + (27.089 \times \text{FD})$	25.891	0.007	0.86
	$\rho_{ap} = 568.884 + (63.928 \times \text{FWT})$	14.044	0.020	0.77
	$v = 8263.277 - (323.280 \times \text{RF})$	9.815	0.035	0.71
<i>Diploptropis</i> spp.				
	$v = 6045.976 - (2.313 \times \text{VEL})$	15.487	0.017	0.79
	$v = 6150.293 - (3.544 \times \text{RH})$	36.142	0.004	0.90
	$v = 5870.813 - (27.282 \times \text{RW})$	11.652	0.027	0.74
	$v = 5728.672 - (262.226 \times \text{FLD})$	42.144	0.003	0.91
	$\text{MOE}_d = 32.929 - (0.0201 \times \text{VEL})$	14.278	0.019	0.78
	$\text{MOE}_d = 33.800 - (0.0307 \times \text{RH})$	29.041	0.006	0.87
	$\text{MOE}_d = 31.488 - (0.240 \times \text{RW})$	11.795	0.026	0.74
	$\text{MOE}_d = 37.978 - (0.612 \times \text{FD})$	7.861	0.049	0.66
	$\text{MOE}_d = 29.818 - (2.165 \times \text{FLD})$	16.929	0.015	0.80
<i>Astronium lecointei</i>				
	$\rho_{ap} = 111.926 + (0.733 \times \text{FL})$	10.969	0.030	0.73
	$\rho_{ap} = 802.494 + (14.179 \times \text{FLD})$	8.969	0.040	0.69
	$v = 2277.071 + (382.027 \times \text{FWT})$	7.756	0.050	0.66

F=statistical significance test, P=significance levels, R²=coefficient of determination. FD=fiber diameter, FL=fiber length, FWT=fiber wall thickness, RH=ray height, RW=ray width, RF=ray frequency, FLD=fiber lumen diameter, VEL=vessel element length, ρ_{ap} =specific gravity, v =speed of sound propagation, MOE_d =dynamic modulus of elasticity.

showed the same trend. The quoted authors studied the non-destructive evaluation of mechanical properties of wood by means of ultrasound. There was a significant correlation among the fibers diameter and lumen diameter and MOE_d in *M. itauba* and *Diploptropis* spp. sticks and between rays dimensions and MOE_d in *Diploptropis* spp. Rao et al. (1997) demonstrated that the different proportion of cells is important for determining wood stiffness. Longui (2005) found higher MOE_d values associated with higher fiber percentages.

It is obvious that a greater similarity with *C. echinata* wood elevates the potential of a wood for bows (model classic-modern). In this study, *Dipteryx* spp., *Handroanthus* spp., and *Hymenaea* spp. fulfill this requirement (Figures 2–8). This result is in agreement with practical experiences in bow manufacture. Professional musicians, including soloists from important Brazilian orchestras, attested high quality to bows made from *Handroanthus* spp. and *Dipteryx* spp. We found that bows made from *Hymenaea* spp. have an adequate qual-

ity to be used by beginners and amateur musicians. *Diploptropis* spp. wood also showed promising characteristics. However, *M. itauba* and *A. lecointei* are not suited for bows.

In practice, wood workability and the acceptance of the musicians are more important than scientific results. Thus, in the production of bows we followed all stages of traditional manufacturing: cutting in band saw, planing, bending with heat, sanding and finishing. Professional bow makers confirmed that the workability of *C. echinata* wood was more difficult than the other six woods studied here with the workability of *M. itauba* and *A. lecointei* being the easiest. Accordingly, an easy workability does not warrant an appropriate quality for bows.

Conclusions

Based on this study and our other investigations (Longui 2005; Alves et al. 2008b; Longui 2009), we recommend the

following indicators for alternative wood species in bow manufacture. It should be emphasized that the recommended data only represent a guideline. The complex wood properties can always influence bow quality in a deviating manner. Wood properties with high potential for good bows: vessel diameter $\approx 110\text{ }\mu\text{m}$; vessel element length $\approx 350\text{ }\mu\text{m}$; vessel frequency $\approx 13\text{ }\mu\text{m}$; ray height $\approx 230\text{ }\mu\text{m}$; ray width $\approx 20\text{ }\mu\text{m}$; ray frequency $\approx 10\text{ }\mu\text{m}$; fiber length $\approx 1160\text{ }\mu\text{m}$; fiber diameter $\approx 18\text{ }\mu\text{m}$; fiber lumen diameter $\approx 5\text{ }\mu\text{m}$; fiber wall thickness $\approx 6\text{ }\mu\text{m}$. Specific gravity $\approx 1000\text{ kg m}^{-3}$; speed of sound propagation $\geq 5300\text{ m s}^{-1}$; dynamic modulus of elasticity $\geq 28\,000\text{ MPa}$.

Acknowledgements

The authors thank Itiberê M.S. Suckow and Diego Romeiro (IF-Forestry Institute) for assistance in the photo preparation and Antonio C.F. Barbosa (IPT-São Paulo Technological Research Foundation) for laboratory assistance.

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Received November 22, 2009. Accepted March 3, 2010.

Previously published online May 18, 2010.